

```

print("Range of resting blood pressure")
max(data$trestbps) - min(data$trestbps)

quantile(data$trestbps, c(0.25, 0.5, 0.75))
print("Column names of the Data")
names(data)
print("Attributes of the Data")
str(data)
print("Number of Rows and Columns:")
dim(data)

[1] "Range of resting blood pressure"
106
25%: 120 50%: 130 75%: 140

[1] "Column names of the Data"
'age' 'sex' 'cp' 'trestbps' 'chol' 'fbs' 'restecg' 'thalach' 'exang' 'oldpeak' 'slope' 'ca' 'thal' 'target'

[1] "Attributes of the Data"
'data.frame': 303 obs. of 14 variables:
 $ age      : int  63 37 41 56 57 57 56 44 52 57 ...
 $ sex      : int  1 1 0 1 0 1 0 1 1 1 ...
 $ cp       : int  3 2 1 1 0 0 1 1 2 2 ...
 $ trestbps : int  145 130 130 120 120 140 140 120 172 150 ...
 $ chol     : int  233 250 204 236 354 192 294 263 199 168 ...
 $ fbs      : int  1 0 0 0 0 0 0 0 1 0 ...
 $ restecg  : int  0 1 0 1 1 1 0 1 1 1 ...
 $ thalach  : int  150 187 172 178 163 148 153 173 162 174 ...
 $ exang    : int  0 0 0 0 1 0 0 0 0 0 ...
 $ oldpeak  : num  2.3 3.5 1.4 0.8 0.6 0.4 1.3 0 0.5 1.6 ...
 $ slope    : int  0 0 2 2 2 1 1 2 2 2 ...
 $ ca       : int  0 0 0 0 0 0 0 0 0 0 ...
 $ thal     : int  1 2 2 2 2 1 2 3 3 2 ...
 $ target   : int  1 1 1 1 1 1 1 1 1 1 ...

[1] "Number of Rows and Columns:"
303 · 14

```

Figure 4: Range of resting blood pressure

Analyze the Correlation between resting BP and age

```

In [7]: print("Correlation between the resting blood pressure and the age")
cor(data$trestbps, data$age, method = "pearson")
cor.test(data$trestbps, data$age, method = "pearson")

[1] "Correlation between the resting blood pressure and the age"
0.279350906561288
Pearson's product-moment correlation

data: data$trestbps and data$age
t = 5.0475, df = 301, p-value = 7.762e-07
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1720897 0.3800657
sample estimates:
cor
0.2793509

```

Figure 5: Correlation between resting BP and age

Constructing the Logistic regression Model

```

print("Constructing the Logistic regression Model")
glm(target~ trestbps+ restecg + fbs, data = data, family=binomial())
model <- glm(target~trestbps+ chol + thalach, data = data, family=binomial())
plot(model)

[1] "Constructing the Logistic regression Model"
Call: glm(formula = target ~ trestbps + restecg + fbs, family = binomial(),
data = data)

Coefficients:
(Intercept)    trestbps    restecg         fbs
  1.98546    -0.01566    0.48160    0.03337

Degrees of Freedom: 302 Total (i.e. Null); 299 Residual
Null Deviance: 417.6
Residual Deviance: 406.6    AIC: 414.6

```

Figure 6: Logistic regression model outcomes

```

print("Maximum resting blood
pressure") max(data$trestbps)
print("Summary of Dataset")
summary(data)

```

To arrive at the range of resting blood pressure, use the following code:

```

print("Range of resting blood
pressure") max(data$trestbps)
- min(data$trestbps)
quantile(data$trestbps, c(0.25, 0.5,
0.75)) print("Column name of the
Data") names(data)
print("Attributes of the Data")
str(data)
print("Number of Rows and Columns:")
dim(data)

```

Figure 4 captures the range of resting blood pressure for the health analysis case study.

To analyse the correlation between resting BP and age, use this code snippet:

```

print("Correlation between the resting
blood pressure and the age")
cor(data$trestbps, data$age, method =
"pearson")
cor.test(data$trestbps, data$age,
method = "pearson")

```

Figure 5 gives the correlation between resting BP and age.

To construct the GLM, type:

```

print("Constructing the Logistic
regression Model")
glm(target~ trestbps+ restecg + fbs,
data = data, family=binomial())
model <- glm(target~trestbps+
chol + thalach, data = data,
family=binomial())
plot(model)

```

The outcomes of the construction of the logistic regression model are illustrated in Figure 6.

Use the following code to construct the decision tree model: